Stage 1
AI as Conversational Assistant

Stage 2
AI as Reactive Operator

Stage 3
AI as Adaptive Coordinator

Stage 4
AI as Self-Governing Production System

Nascent Diffusion Industries:

Limited AI penetration · efficiency-driven assistance · human-supervised

Manufacturing Industry

Applications

- Defect Detection
- Engineering & Design Assistance
- Operational Decision Support

Current Limitations

- **System Complexity:** Struggles to maintain state across systems.
- **Safety Constraints:** Restricted by stringent physical safety requirements and high downtime costs.
- **Strict Governance:** Requires tight human oversight.

Agriculture

Applications

- Plant Disease Diagnosis
- Agricultural Decision Support
- Cross-validation of Consultation

Current Limitations

- **"Consultant" Phase:** Lack execution capabilities.
- **Edge Instability:** Small models (4B) struggle with complex in rural settings.
- **Safety Risks:** Potential for misleading.
- **Resource Gaps:** Between limited computing power and data privacy.

Real Estate

Applications

- Property Price Estimation
- Transaction Information Extraction
- Property Marketing Copywriting

Current Limitations

- **Limited Autonomy:** Remain reactive, lacking long-horizon planning.
- **Hallucination Risk:** External methods are needed to obtain reliable facts.
- **Static Workflows.**
- **Poor Numerical Reasoning:** Poor on numerical valuation tasks.

Government & Pub. Admin.

Applications

- Administrative Business Consultation
- Social Stimulation
- Administrative Document Writing

Current Limitations

- **Passive Execution:** Lack the ability to orchestrate workflows.
- **Execution Gap:** Cannot perform real-world administrative actions.
- **Supervised Accountability.**
- **Infrastructure Barriers:** Absence of modern APIs on public sector websites.